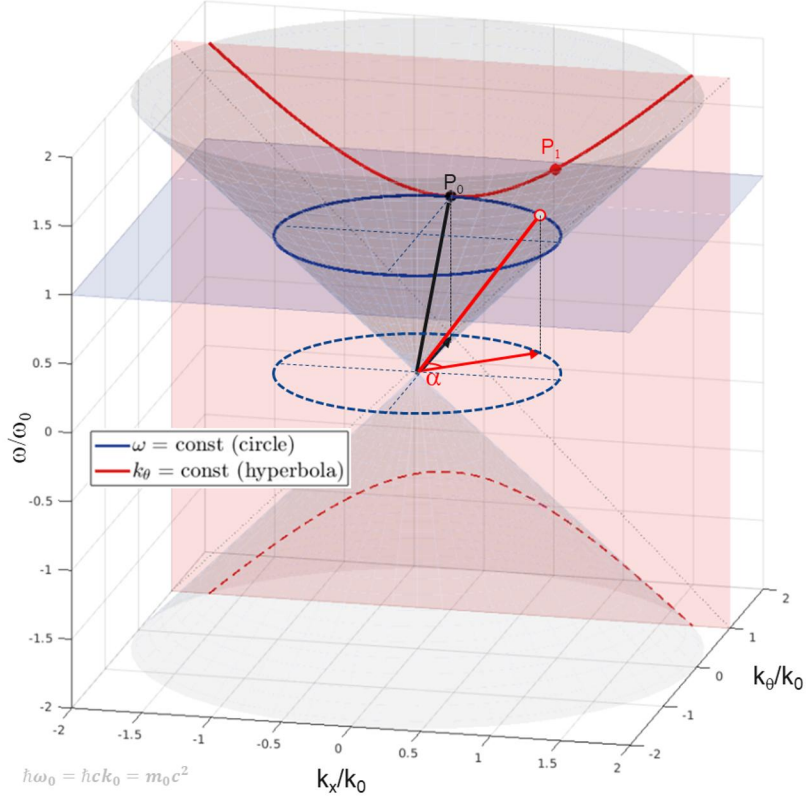
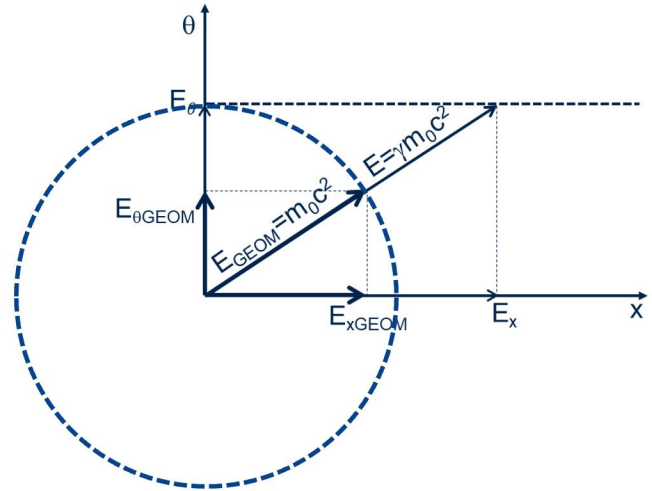
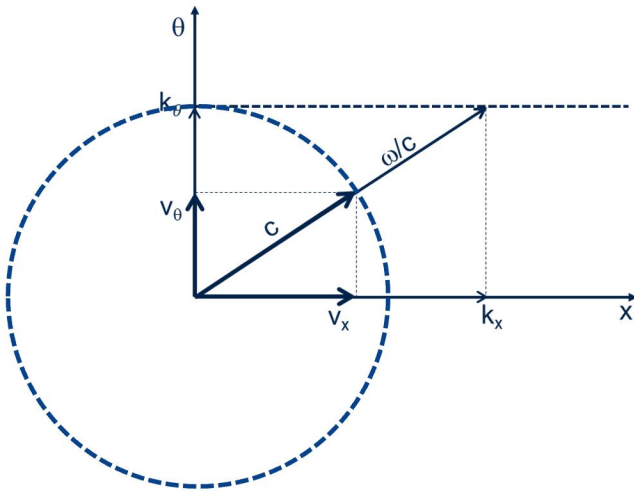
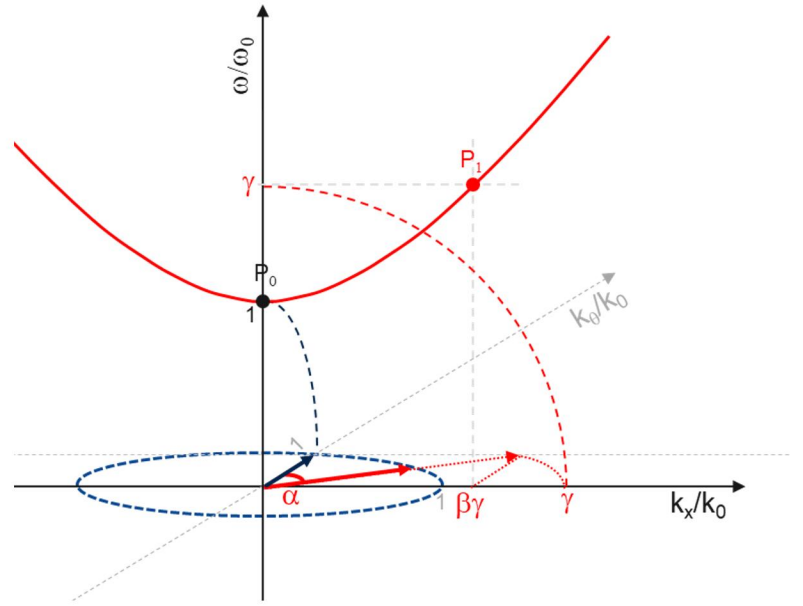


Dispersion cone $\omega^2 = c^2(k_x^2 + k_\theta^2)$
circle (ω const) vs hyperbola (k_θ const)



Circle-Hyperbola geometric relation



$$\omega^2 = c^2(k_\theta^2 + k_x^2)$$
$$k_\theta = \frac{2\pi}{\lambda_0} = \frac{m_0 c}{\hbar} \quad \text{invariant}$$

$$v_{\theta}^2 = \left(\frac{\delta \omega}{\delta k_{\theta}} \right)^2 = \frac{k_{\theta}^2}{k_{\theta}^2 + k_x^2} c^2 = \frac{k_{\theta}^2}{\omega^2} c^4$$

$$v_x^2 = \left(\frac{\delta \omega}{\delta k_x} \right)^2 = \frac{k_x^2}{k_\theta^2 + k_x^2} c^2 = \frac{k_x^2}{\omega^2} c^4$$

$$c^2 = v_\theta^2 + v_x^2$$

c invariant

$$E^2 = E_\theta^2 + E_x^2$$

$$E_\theta = m_0 c^2 \quad \text{invariant}$$

$$E_x = p_x c$$

$$E = \gamma m_0 c^2$$

$$E_{xGEOM} = \beta m_0 c^2$$

$$E_{GEOM} = m_0 c^2$$

$$E_{GEOM}^2 = E_{\theta GEOM}^2 + E_{xGEOM}^2$$

$$E_{GEOM} = m_0 c^2 \quad \text{invariant}$$

Hyperbola

Circle